RUNGTA INTERNATIONAL SKILLS UNIVERSITY

SCHOOL OF INFORMATION TECHNOLOGY
BACHELOR OF COMPUTER APPLICATION

TEST CASES - PS No. 25: Menu-Driven Calculator

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Course	BCA (AI and ML)	**Section**	1-B

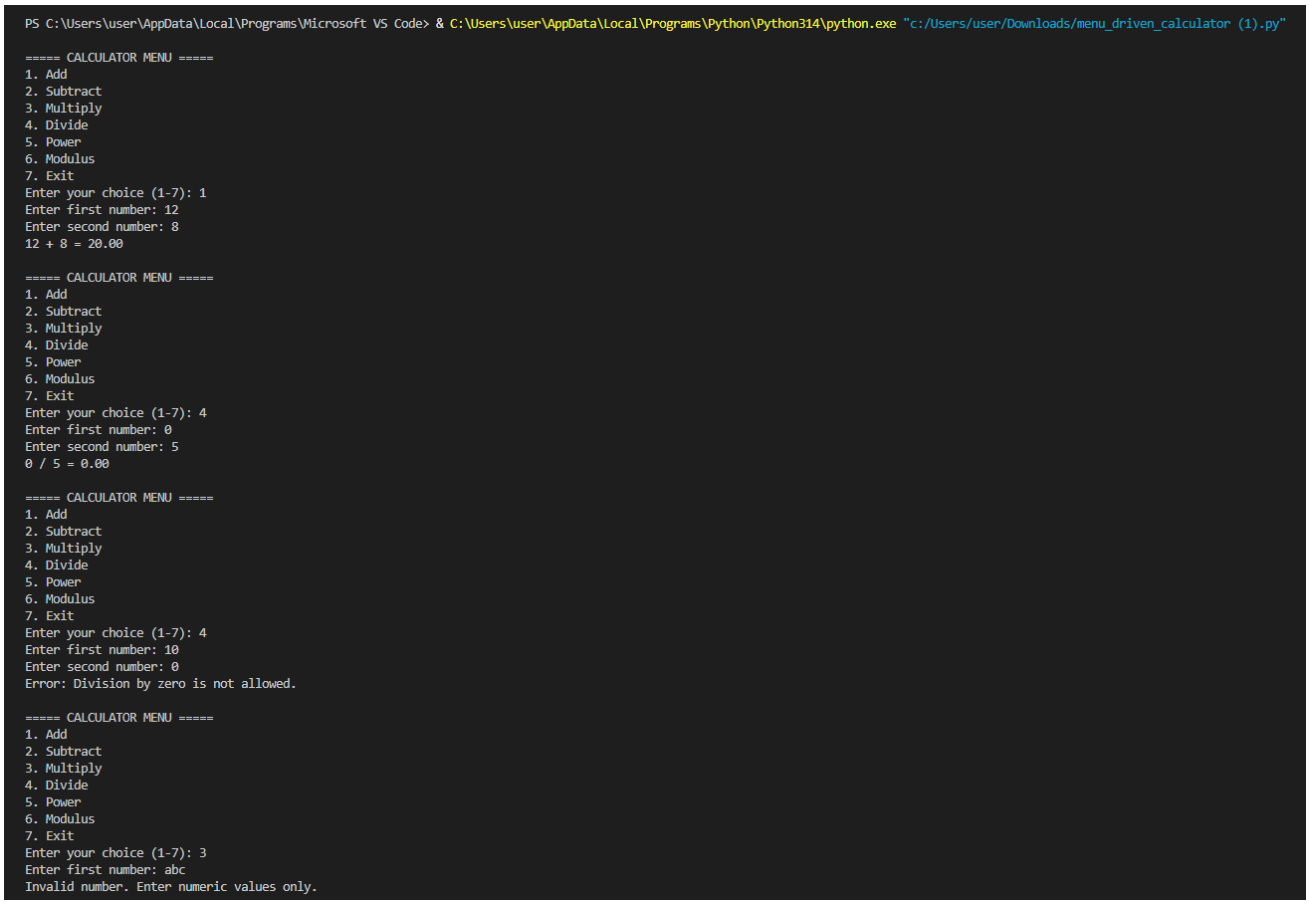
Summary of Test Cases

Test Case	Input	Expected Output	Result
1. Normal Input	Option 1 (Add): 12 and 8	12 + 8 = 20.00	**PASS**
2. Boundary / Minimum Input	Option 4 (Divide): 0 and 5 (zero numerator)	0 / 5 = 0.00	**PASS**
3. Invalid Input	(a) Option 4 (Divide): 10 and 0 (b) Option 3 (Multiply): abc and 4 (c) Menu choice: 9	(a) Error: Division by zero is not allowed. (b) Invalid number. Enter numeric values only. (c) Invalid choice. Select 1 to 7.	**PASS**
4. Different Valid Scenario	(a) Option 5 (Power): 2 and 10 (b) Option 6 (Modulus): 17 and 5 (c) Option 2 (Subtract): 5 and 9.5 (d) Option 7 (Exit)	(a) 2 ^ 10 = 1024.00 (b) 17 % 5 = 2.00 (c) 5 - 9.5 = -4.50 (d) Thank you for using the calculator. Exiting...	**PASS**

The screenshots on the following pages show the program running in VS Code (Python 3.14) and match every expected output above.

Screenshot 1 - Test Case 1, Test Case 2, Test Case 3(a) and 3(b)

Add 12 and 8 gives 12 + 8 = 20.00; Divide 0 and 5 gives 0 / 5 = 0.00; Divide 10 and 0 gives the division-by-zero error; Multiply abc and 4 gives the invalid number message.



Screenshot 2 - Test Case 3(c), Test Case 4(a) and 4(b)

Menu choice 9 gives Invalid choice. Select 1 to 7.; Power 2 and 10 gives 2 ^ 10 = 1024.00; Modulus 17 and 5 gives 17 % 5 = 2.00.

```
===== CALCULATOR MENU =====
1. Add
2. Subtract
3. Multiply
4. Divide
5. Power
6. Modulus
7. Exit
Enter your choice (1-7): C:\Users\user\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/user/Downloads/menu_driven_calculator (1).py"
Invalid choice. Select 1 to 7.

===== CALCULATOR MENU =====
1. Add
2. Subtract
3. Multiply
4. Divide
5. Power
6. Modulus
7. Exit
Enter your choice (1-7): 9
Invalid choice. Select 1 to 7.

===== CALCULATOR MENU =====
1. Add
2. Subtract
3. Multiply
4. Divide
5. Power
6. Modulus
7. Exit
Enter your choice (1-7): 5
Enter first number: 2
Enter second number: 10
2 ^ 10 = 1024.00

===== CALCULATOR MENU =====
1. Add
2. Subtract
3. Multiply
4. Divide
5. Power
6. Modulus
7. Exit
Enter your choice (1-7): 6
Enter first number: 17
Enter second number: 5
17 % 5 = 2.00

===== CALCULATOR MENU =====
1. Add
2. Subtract
3. Multiply
```

Screenshot 3 - Test Case 4(c) and 4(d)

Subtract 5 and 9.5 gives 5 - 9.5 = -4.50; Option 7 prints the thank-you message and exits.

